

Let's Q (Q 一下)

Official App Development & Strategy Document — Revision 4 (Complete)

Consolidates every decision made through the full design conversation — formerly Mello Blitz Queue

1. Executive summary (執行摘要)

Let's Q (Q 一下) is a single cross-platform app, used by two roles: anonymous queuers, who scan a QR code and hold a place in line with no name, phone number, or email, and event organizers, who run one or more queues per booth from a real-time console. There is no separate app for each role — the same download works for both, since queuers may later become organizers themselves.

2. Global accessibility (多語言支援)

Six languages are supported at launch: Traditional Chinese, Simplified Chinese, English, Spanish, Japanese, and Korean. A queuer selects their language together with their personal ticket color, both required before joining a queue. An organizer selects a primary language for the printed QR sheet at queue setup; the digital ticket view still follows each queuer's own phone language regardless.

3. The organizer ecosystem (活動方管理端)

- **Sign-in is optional:** an organizer can start immediately with no sign-in — the \$1/month subscription bills automatically through the phone's existing App Store/Play Store account. Linking an Apple or Google account is offered separately, for recovering an event, its analytics, and saved templates from another device.
- **Required first-time setup, before the Dashboard unlocks:** every path (no sign-in, Apple, or Google) leads to one required setup screen, not straight to the Dashboard. It must be completed in full before "Publish queue" activates: booth/event name or photo, the first queue's name, its color, the primary print language, capacity, and the no-show policy.
- **Up to 2 queues per booth:** e.g. a "Photo queue" and a "Prize pickup" queue under one booth name. Each queue has its own number sequence, capacity, no-show policy, and QR code.
- **Capacity control:** a preset limit can be adjusted live in either direction. Lowering capacity never revokes tickets already issued — it only limits new scans until the queue drops back under the new limit.
- **Pause / resume:** pausing blocks new QR scans only; customers already in the queue keep moving and being called normally.
- **Queue-full waitlist:** when a queue is at capacity, a queuer scanning in is offered a waitlist instead of being blocked outright, and is notified the moment a spot opens.
- **Early closure:** an organizer closing a queue before it finishes must type a message, which is broadcast immediately to every remaining ticket holder's screen.
- **Staff device pairing:** a QR code (primary, single-use, expires in 10 minutes) or a PIN (fallback, locks after 5 wrong attempts) lets a booth device run the queue without needing the owner's own Apple/Google login.
- **Walk-in tickets:** staff can manually create a ticket on their own device for a visitor without a smartphone; it counts toward the same capacity and no-show rules as any other ticket.
- **Optional phone number:** an organizer can choose to share a contact phone number, shown on the queuer's waiting ticket if enabled.
- **Queue color:** required at queue setup — this is the color shown on that queue's row in every queuer's "My Queues" list.

4. The user experience (排隊者使用端)

- **The secret code:** instead of a name, phone number, or email, a queuer creates a short code after scanning the QR code. It is never spoken aloud — it is shown on screen and checked visually by the organizer or the queuer. The system rejects a code already active in the same queue.
- **Optional message to the booth:** up to 30 characters for Chinese/Japanese/Korean, or 60 characters for English/Spanish, shown to the organizer ahead of time so they can prepare (e.g. an accessibility need).
- **One ticket per queue at a time:** a queuer can hold only one active ticket per queue; once that ticket is completed or cancelled, they are free to join again.
- **"My Queues" list:** shows every active ticket across different booths at once, each card colored by that queue's own organizer-chosen color, so 3 or more simultaneous tickets are easy to tell apart.
- **Self-cancellation:** a queuer can cancel their own ticket at any time; the number is marked cancelled/vacant and is never reassigned or renumbered.
- **Share this ticket:** a queuer can hand off a ticket's live status and secret code to someone else via QR or link — useful when two people are queueing together and one needs to leave early.
- **3-question satisfaction survey:** after being served, one tap each on "How was your wait?", "How was the service?", and "Will you come again?", each on a 1–5 face-icon scale. Fully anonymous, feeding directly into the organizer's post-event report.

5. No-show & queue management rules (叫號規則)

Set once per queue, before it opens, and printed on the QR sheet plus shown on the queuer's ticket:

Policy	How it works
Cancel immediately	First no-show cancels the ticket instantly; the queuer is notified in-app to re-join with a new ticket.
Defer 5 spots, twice, then cancel	No-show pushes the number back 5 spots and recalls it; after 2 deferrals (3 total call attempts), it auto-cancels.
Hold N minutes, twice, then cancel	No-show holds the number for N minutes (organizer-set) and recalls it; after 2 holds (3 total call attempts), it auto-cancels.

On the dashboard, calling a number shows three organizer actions: Present (queuer showed up, ticket completed normally), Skip (no penalty, manual hold, recall anytime — useful for elderly or slower queuers), and No-show (automatically runs the queue's preset policy above, no manual decision needed in the moment).

6. UI/UX design & customization (美編與介面自訂)

The app uses one fixed, classic layout across both roles — not multiple templates — with a 7-color accent palette (red, orange, yellow, green, blue, purple, black) providing visual variety without adding setup friction or multiplying the QA surface across templates.

- **Organizer's queue color:** required when creating the queue; this is what tints that queue's row in the queuer's "My Queues" list.
- **Queuer's personal color:** required alongside language selection, right before joining a queue; tints only that queuer's own ticket screens.
- **Fixed status colors:** the amber "waiting" and green "your turn" signal colors never change regardless of theme, for accessibility.

7. Privacy, security & compliance logic (隱私、安全與合規邏輯)

- **Scope of the privacy claim:** the queue itself never stores names, phone numbers, or emails, so the queue data and post-event analytics carry no GDPR/PII exposure.
- **Ad SDK disclosure:** Google Mobile Ads reads the device's advertising identifier for ad targeting, which requires an App Tracking Transparency (ATT) prompt on iOS and a consent notice for GDPR/CCPA-covered users — a separate obligation from the queue data itself.
- **Anti-abuse controls:** one active ticket per device token per queue, preventing queue spam from repeated scans.

8. Monetization & ad strategy (商業模式與廣告策略)

Ad type	Queuer	Organizer
Full-page, on app open (max 10s)	Yes	Yes
Full-page, every 10 min any screen (max 10s)	Yes	Yes, paused while actively calling numbers
Banner, on screen	Yes, always	Yes, always — even for subscribers
\$1/month subscription removes	Nothing — unaffected	Both full-page ad types only

The app remains free to download for both roles. The \$1/month organizer subscription removes only the organizer's own full-page ads (on app open and the 10-minute recurring one); it has no effect on queuer ads, and banner ads persist for everyone regardless of subscription status, since the free download itself is subsidized by banner ad revenue.

9. Post-event analytics (活動後資料分析)

- **Core metrics:** total queued, average wait time, no-show rate, and peak-hour traffic — aggregate counts and timings only, no personal data.
- **Queue-vs-queue comparison:** for booths running 2 queues, wait times are compared side by side to show which needs more staffing.
- **Capacity utilization:** flags time windows where the queue was at its limit, quantifying how many people likely couldn't join.
- **Satisfaction breakdown:** average score and distribution for each of the 3 survey questions (wait, service, would-return), highlighting when they diverge (e.g. service scoring lower than wait time points to a staffing issue, not a queue-speed issue).
- **Actionable suggestions:** the report translates the numbers into a concrete recommendation for the next event (e.g. capacity and staffing changes), rather than only presenting raw statistics.

10. Technical stack & deployment (技術與部署架構)

Component	Version / detail
Frontend	Flutter 3.44, Dart 3.12
Android target	API 36 (Android 16); Google Play requires this by Aug 31, 2026
Java compatibility	Java 17 (not the deprecated Java 1.8 flag, which risked D8/R8 dexing)

	conflicts)
iOS	Xcode 26 / iOS 26, developed on macOS Tahoe
Backend	Firebase Cloud Firestore, real-time sync
Ads	Google Mobile Ads SDK, with ATT (iOS) and GDPR/CCPA consent handling
Billing	Apple StoreKit / Google Play Billing, tied to the device's existing Store account — no separate login required to subscribe

11. Revision log — everything added since Revision 3

#	Change since Revision 3
1	Added multi-queue-per-booth (max 2), each with its own number sequence, capacity, and QR code, sharing one booth name.
2	Added the queuer's "My Queues" list, with each card color-coded by that queue's organizer-chosen color — the mechanism that lets someone holding several tickets tell them apart.
3	Added an optional message field (queuer to organizer) with a language-based character limit: 30 for CJK languages, 60 for Latin-script languages.
4	Replaced ad-hoc no-show handling with 3 explicit, configurable policies, set once per queue and printed/displayed wherever the ticket appears.
5	Added Present / Skip / No-show as the organizer's three call-time actions; No-show triggers the preset policy automatically.
6	Defined ticket self-cancellation: number is marked cancelled/vacant, never renumbered or reassigned.
7	Added a queue-full waitlist: when a queue is at capacity, the queuer can join a waitlist and is notified when a spot opens, instead of being blocked outright.
8	Added mandatory-message early closure: an organizer closing a queue early must type a message, broadcast immediately to all remaining ticket holders.
9	Added staff device pairing (QR primary, PIN fallback) so booth staff can run a queue without using the owner's Apple/Google account.
10	Added staff-assisted walk-in ticket creation for visitors without a smartphone; counts toward the same capacity and no-show rules.
11	Made organizer sign-in optional: "Start now, no sign-in" is the default (billing works via the device's Store account automatically); linking an Apple/Google account is offered separately for cross-device recovery.
12	Added an optional organizer phone-number toggle, shown on the queuer's waiting ticket if enabled.
13	Locked the visual design to one fixed layout with a 7-color accent palette (red/orange/yellow/green/blue/purple/black), chosen instead of multiple template layouts — color solves the actual need (telling tickets apart) without adding setup friction or QA surface.
14	Defined two separate, required color choices: the organizer's queue color (set at queue creation, colors that queue's row in every queuer's My Queues list) and the queuer's personal ticket color (chosen alongside language, right before joining, colors only their own ticket screens).
15	Confirmed the amber "waiting" / green "your turn" status colors never change regardless of any theme

	choice, for accessibility.
16	Added a "Share this ticket" feature: a queuer can hand off their ticket's live status and secret code to someone else via QR or link (e.g. a partner leaving early).
17	Expanded the satisfaction pulse into a 3-question survey (wait, service, would-return), each on a 1–5 face-icon scale, one tap per question, fully anonymous.
18	Revised the ad strategy: full-page ads (max 10s) now appear on app open and every 10 minutes for both queuers and organizers (paused for organizers while actively calling); banner ads are permanent for everyone since the app is free; the \$1/month subscription removes only the organizer's own full-page ads.
19	Locked language support at 6 for v1 (English, Spanish, Traditional Chinese, Simplified Chinese, Japanese, Korean), selected together with the queuer's color choice; organizer selects a primary print language at queue setup.
20	Updated the technical stack to current versions: Flutter 3.44/Dart 3.12, Android API 36, Java 17, Xcode 26/iOS 26.
21	Decision log from stress-testing a 1,000-person / 10-booth outdoor event: kept booths fully independent (no event-level layer above booths); kept ad frequency uniform regardless of how many queues a person is in.
22	Added a required first-time setup screen between login and the Dashboard — booth/event name or photo, first queue name, color, print language, capacity, and no-show policy must all be completed before the Dashboard becomes accessible.